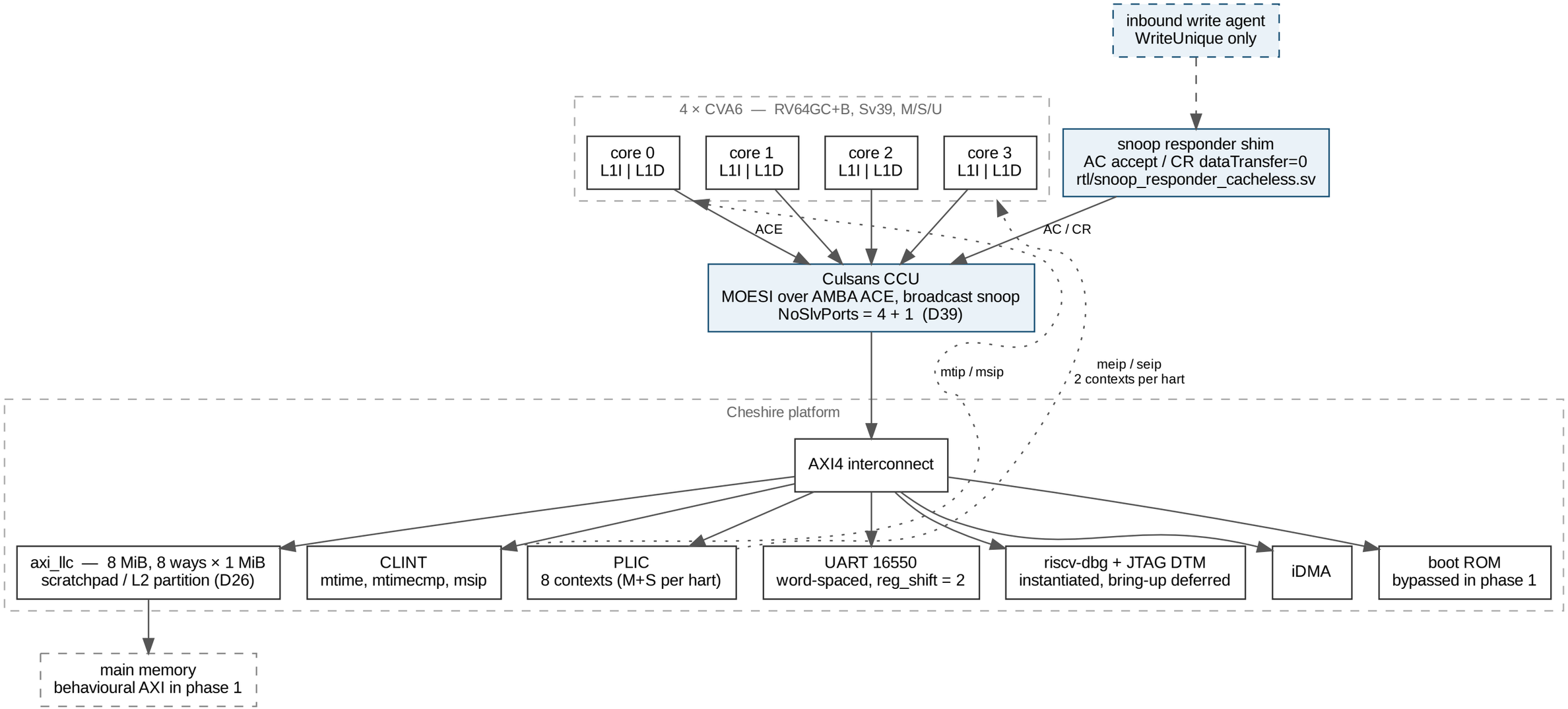
D1 — cva6-multi cluster internals
inbound write agent
WriteUnique only
4 × CVA6 — RV64GC+B, Sv39, M/S/U
core 0
L1I | L1D
core 1
L1I | L1D
core 2
L1I | L1D
core 3
L1I | L1D
snoop responder shim
AC accept / CR dataTransfer=0
rtl/snoop_responder_cacheless.sv
ACE
AC / CR
Culsans CCU
MOESI over AMBA ACE, broadcast snoop
NoSlvPorts = 4 + 1 (D39)
mtip / msip
meip / seip
2 contexts per hart
Cheshire platform
AXI4 interconnect
axi_llc — 8 MiB, 8 ways × 1 MiB
scratchpad / L2 partition (D26)
CLINT
mtime, mtimecmp, msip
PLIC
8 contexts (M+S per hart)
UART 16550
word-spaced, reg_shift = 2
riscv-dbg + JTAG DTM
instantiated, bring-up deferred
iDMA
boot ROM
bypassed in phase 1
main memory
behavioural AXI in phase 1